Guide to Oscillations, Resonance & Noise

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November 10, 2026

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A small collection of interactive animations and one Jupyter notebook about oscillations and resonance: from a free spring, through damping, forcing and resonance, to a particle in a double-well potential driven by noise (stochastic resonance). Everything is plain HTML, with no installation and no build step.

Live site: https://diegomaragnano.github.io/resonance/

Guide to the animations

Most animations have an animated scene (spring or potential) and one or more plots. Sliders update the simulation immediately.

1 Free spring-mass system

Unforced and undamped motion.

- **Drag the mass** up or down: sets the amplitude x_0.
- **Set f_0 directly / Set k and m**: choose how the natural frequency is defined.
  - **f_0 slider (0.25–2 Hz)**: sets the frequency directly.
  - **m slider (0.5–2 kg) and k slider (10–75 N/m)**: set mass and spring constant; the frequency follows from them.
- **Pause / Reset**: stop and resume the motion; restore the initial state.
- **Time in s / Time in periods**: unit of the time axis of the plot.
- **Energy bars**: shows a panel with kinetic and potential energy.
- **Phase portrait**: shows a panel with velocity against position.
- An information box reports the current parameters: the angular frequency ω_0, the frequency f_0, and the period T.

2 Damped spring-mass system

Equation $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$, damping ratio $\zeta = \gamma/\omega_0$.

- Drag the handle on the mass to choose a new starting point.
- ω_0 slider (1–6 rad/s): natural frequency. γ slider (0–8 s^{-1}): damping rate.
- Under-damped / Critical / Over-damped: presets that set γ to a typical value for each regime. A badge and an information box show the current regime, and quantities that are relevant for that regime.
- Release from $x = +1$ m: restarts the motion from a fixed initial displacement.
- Pause and speed menu (0.25×, 0.5×, 1×).

- Check-boxes:
  - Ghost trail: leaves fading copies of the mass.
  - Equilibrium line: marks x = 0.
  - Envelope ±: decay envelope (available only in the under-damped regime).
  - Settling band ±2%: band within which the motion is considered settled.
  - Damping force: shows the force from the damper.
  - Time in periods T_0: time axis in units of the natural period.
  - Phase portrait: shows velocity against position.
  - Restart on parameter change: if on, moving a slider restarts the motion; if off, the current position and velocity are kept and the motion continues with the new parameters.
- The colour of the fluid in the damper indicates the damping strength (low to high γ).

3 Forced undamped system

Equation $m\ddot{x} + kx = F_0 cos(\omega t)$, with $m = 1$ kg, $k = 4$ N/m (so $\omega_0 = 2$ rad/s), starting at rest.

- Pause/Play and Restart.
- Speed menu: 1×, 2×, 4×.
- ω/ω_0 slider (0.5–1.5): ratio of the driving frequency to the natural one. Near 1 you see resonance (steady growth); slightly away from 1, beating.
- F_0 slider (0.2–1.2 N): amplitude of the driving force.
- Check-boxes:
  - Envelope: draws the amplitude envelope.
  - Support motion $y(t)$: shows how the support moves (exaggerated in the scene).
  - Phase portrait $x^{\smile}\dot{x}$: shows velocity against position.
  - Auto-restart: restarts the run automatically after it ends.
- A status line describes what is happening.

4 Resonance lab

Driven, damped oscillator with several linked views: the animated spring, the position trace, the resonance curve and a readout panel.

- Drive frequency slider: ratio ω/ω_0 (0.2–3).

- Quality factor Q slider: sets the damping (high Q = light damping). The fluid colour goes from amber (heavy damping) to blue (light damping).
- Drive amplitude slider.
- Drive type: Moving support or Force on the mass.
- Stiffness slider: natural frequency $\omega_0 = \sqrt{k/m}$, with $m = 1$.
- Speed menu: 0.25× to 16×.
- Buttons:
  - Snap to $\omega = \omega_0$: jumps exactly to resonance.
  - $\omega/\omega_0 = 0.5$ and $\omega/\omega_0 = 2$: jump to those ratios.
  - Sweep through resonance: automatically sweeps the drive frequency across the resonance.
  - Stop drive: switches the driving off so the system decays freely.
  - Pause, Reset.
  - Phase portrait: switches the position panel to a velocity-versus-position view.
  - Phase panel: shows or hides a plot of the phase lag against frequency ratio.
- The readout panel displays the main quantities, including Q.

5 Marble in a double well

Dynamics $\dot{x} = x(a - x^2) = -V'(x)$, with $V(x) = x^4/4 - ax^2/2$.

- Click or drag on the potential to drop the marble at that position.
- a slider (-1 to 3): changes the shape of the potential. For $a \leq 0$ there is a single well; for $a > 0$ two wells appear, with the central point becoming unstable.
- Pause/Play and Reset.
- The shading marks the basins of attraction; the second plot shows $x(t)$ with dashed lines at the fixed points. A readout reports the current time, position and destination.

6 Langevin marble

The same marble with noise: $dx = x(a - x^2)dt + \varepsilon dW$.

- a slider (-2 to 3): shape of the potential. ε slider (0.05-1.5): noise strength. Speed slider (1-40).
- Pause/Resume, Reset marble, Clear histogram.
- Trail: shows the recent path of the marble.

- Colour bands in $x(t)$: shades the trajectory plot according to which well the marble is in.
- The scene has two bulbs under the wells: the lit one is the well the marble currently occupies.
- Plots: the potential with the marble, the live trajectory $x(t)$ (last 100 time units), and a histogram of where the marble has spent its time, compared with the exact stationary density (red curve). The counter shows time, position and number of jumps between wells.

7 Stochastic resonance

Rocking double well with noise (Benzi, Sutera, & Vulpiani, 1981): $dx = [x(a - x^2) + A\cos(\Omega t)]dt + \varepsilon dW$

- Sliders: a (well shape), A (forcing amplitude), Ω (forcing frequency), ε (noise), speed, and tilt display (changes only how strongly the tilt is drawn, not the dynamics).
- Pause, Reset, and Colour bands in $x(t)$.
- Presets: too little noise, resonance and too much noise set ε relative to the optimal value.
- The information panel shows: the threshold forcing A_c (below it the marble cannot cross without noise), the mean exit time compared with half the forcing period, the paper's resonance window for ε, the optimal noise level, and the lock-in response (how much of the ideal switching amplitude the marble achieves), with a bar.
- Plots: the tilted potential with the marble and a gold marker of the forcing phase; $x(t)$ over the last three periods with the signal as a dashed green line; a histogram of residence times between jumps, in units of half a period.

Notebook

`Plots notebook` shows the content of `notebooks/plots.html`, which contains the code and figures: free, damped, beating, forced and resonant oscillators, Langevin dynamics, the tilted double well and the stochastic simulation. The text of the notebook is in Italian. The original `.ipynb` can be downloaded from the index.